IBM APPLIED DATA SCIENCE CAPSTONE

# Data Science Capstone Project Report

Predicting whether the SpaceX Falcon 9 first stage will land

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**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone

# Project GitHub Repository

**Repository:** https://github.com/khoalee252006/spacex-falcon9-capstone

| | | |
|---|---|---|
| 1 | **Data Collection – SpaceX API** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/01_data_collection_api.ipynb |
| 2 | **Data Collection – Web Scraping** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/02_data_collection_webscraping.ipynb |
| 3 | **Data Wrangling** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/03_data_wrangling.ipynb |
| 4 | **EDA with Data Visualization** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb |
| 5 | **EDA with SQL** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb |
| 6 | **Interactive Map with Folium** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb |
| 7 | **Plotly Dash Dashboard** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/07_dash_app.py |
| 8 | **Predictive Analysis** | https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb |

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**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone

# Executive Summary

## Methods

- Data collection: SpaceX REST API + web scraping (Wikipedia)
- Data wrangling: missing values, landing label Class (1 / 0), one-hot encoding
- EDA: pandas, matplotlib / seaborn charts and SQL (SQLite)
- Interactive analytics: Folium map and Plotly Dash dashboard
- Machine learning: Logistic Regression, SVM, Decision Tree, KNN with GridSearchCV (cv = 10)

## Key results

- 90 Falcon 9 launches (2010–2020); 60 landed = 66.7% success
- Success grew from 0% (2010–2013) to 90% in 2019
- Best common orbit: VLEO 86%; weakest: GTO 52%
- When a landing was attempted, 87.0% succeeded
- Logistic Regression, SVM, KNN: 83.3% test accuracy (tie); SVM has the best validation accuracy (84.8%)

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone

# Introduction

## Background

- SpaceX advertises Falcon 9 launches at about $62 million.
- Other providers charge $165 million or more per launch.
- Most of the saving comes from landing and reusing the first stage.
- A new company ("Space Y") wants to bid against SpaceX.

## Problem statement

- Can we predict if the Falcon 9 first stage will land successfully?
- If yes, we can estimate the real cost of a launch.

## Questions we answer

- How do payload mass, orbit, launch site and flight number affect landing?
- Did landing success improve over time?
- Where are the launch sites, and why there?
- Which machine-learning model predicts landing best?

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone

METHODOLOGY

# Data Collection – SpaceX API




## Key phrases

- requests.get() → JSON response
- pd.json_normalize() to flatten the JSON
- Helper look-ups: BoosterVersion, LaunchSite, PayloadMass, Orbit, Outcome, Block, Serial…
- Missing PayloadMass (5) → filled with mean (6,123.55 kg)

## Note on data source

- The live API returned HTTP 522/525 errors (server down) in Oct 2026.
- The code tries the live API first, then falls back to saved copies of the same API responses: IBM's static launch JSON + archived responses (web.archive.org).
- Check: 15 of 17 columns are identical to IBM's reference dataset_part_1.csv.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/01_data_collection_api.ipynb

METHODOLOGY

# Data Collection – Web Scraping




## Key phrases

- Page: "List of Falcon 9 and Falcon Heavy launches" – fixed old revision used by the IBM lab (June 2021), so results repeat
- 9 launch tables found; flight number is in the <th> cell, other values in <td> cells
- Columns: Flight No., Date, Time, Version Booster, Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, Booster landing
- Cleaning: remove footnote marks like [12], parse "4,700 kg" → 4700, parse dates
- Result: 121 launches from 04 Jun 2010 to 06 Jun 2021

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/02_data_collection_webscraping.ipynb

METHODOLOGY

# Data Wrangling




- PayloadMass: 5 missing → replaced by mean
- LandingPad: 26 missing (28.9%) → kept, means "no landing pad used"
- Good outcomes (Class = 1): True ASDS (drone ship), True RTLS (ground pad), True Ocean
- Bad outcomes (Class = 0): False ASDS / RTLS / Ocean, None ASDS, None None
- **Success rate = mean(Class) = 66.7%**
- Output: dataset_part_2.csv (+ Class) and dataset_part_3.csv (80 numeric features)

| Outcome | Launches | Class |
|---|---|---|
| True ASDS | 41 | 1 |
| None None | 19 | 0 |
| True RTLS | 14 | 1 |
| False ASDS | 6 | 0 |
| True Ocean | 5 | 1 |
| False Ocean | 2 | 0 |
| None ASDS | 2 | 0 |
| False RTLS | 1 | 0 |

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/03_data_wrangling.ipynb

METHODOLOGY

# EDA with Data Visualization

| Chart type | Chart | Why we use it |
|---|---|---|
| Scatter | Flight Number vs. Payload Mass | Does experience or payload change landing? |
| Scatter | Flight Number vs. Launch Site | Do sites improve over time? |
| Scatter | Payload Mass vs. Launch Site | Which site handles heavy payloads? |
| Bar | Success Rate by Orbit | Which orbit is easiest to land from? |
| Scatter | Flight Number vs. Orbit | Does each orbit improve with flights? |
| Scatter | Payload Mass vs. Orbit | Is heavy payload a problem in some orbits? |
| Line | Yearly Success Trend | Is SpaceX getting better each year? |

Points are colored by Class: green = landed, red = not landed. Tools: pandas, matplotlib, seaborn.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

METHODOLOGY

# EDA with SQL

| # | Question | Key SQL |
|---|---|---|
| 1 | Unique launch sites | SELECT DISTINCT |
| 2 | 5 records with site starting 'CCA' | LIKE 'CCA%' LIMIT 5 |
| 3 | Total payload for NASA (CRS) | SUM() |
| 4 | Average payload of F9 v1.1 | AVG() |
| 5 | First successful ground-pad landing | MIN(Date) |
| 6 | Drone-ship success, payload 4000–6000 kg | AND, > < |
| 7 | Count of mission successes / failures | COUNT(*) GROUP BY |
| 8 | Boosters with maximum payload | subquery MAX() |
| 9 | 2015 drone-ship failures by month | substr(Date) |
| 10 | Rank landing outcomes 2010-06-04 → 2017-03-20 | BETWEEN, ORDER BY DESC |

Data: IBM course table Spacex.csv (101 Falcon 9 launches) loaded into SQLite with pandas.to_sql().

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

METHODOLOGY

# Interactive Visual Analytics – Folium & Plotly Dash

## Folium map

- folium.Circle + Marker: each launch site
- MarkerCluster: green = landed, red = not landed
- MousePosition: read coordinates on the map
- PolyLine + distance label: site → coast, railway, highway, city
- Nearest features found with OpenStreetMap (Overpass API) + haversine distance

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

## Plotly Dash dashboard

- Dropdown: All sites or one launch site
- Pie chart: successful launches by site / success vs. failure
- RangeSlider: choose a payload range (kg)
- Scatter: payload vs. launch outcome, colored by booster version
- Callbacks update the charts when the user changes inputs

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/07_dash_app.py

# Predictive Analysis – Methodology

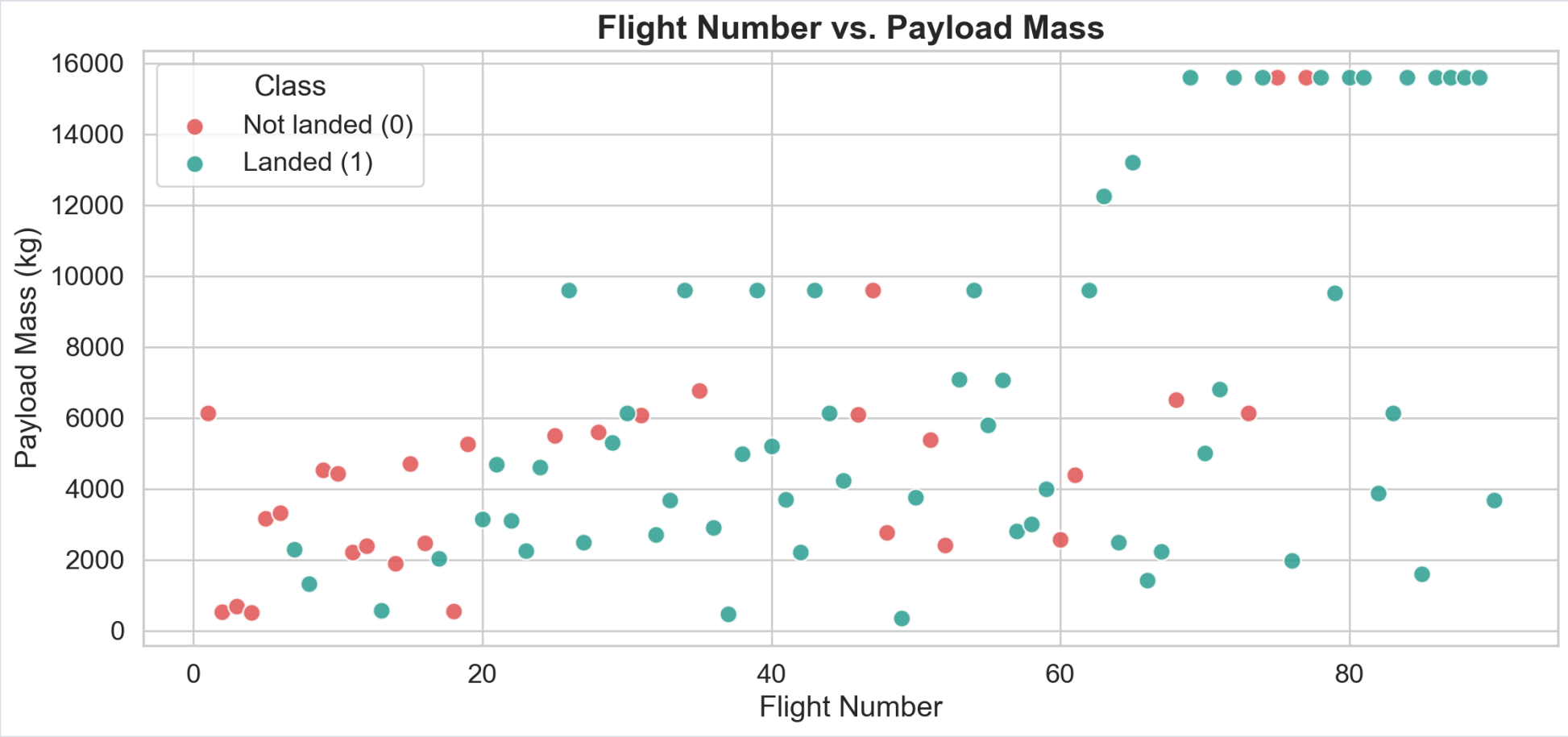



| Model | Parameters searched |
|---|---|
| Logistic Regression | C = 0.01, 0.1, 1 |
| SVM | kernel = linear / rbf / poly / sigmoid; C, gamma = logspace(-3, 3, 5) |
| Decision Tree | criterion, splitter, max_depth 2–18, max_features, min_samples_leaf / split |
| KNN | n_neighbors 1–10, algorithm, p = 1 / 2 |

Extra check: the test set has only 18 launches, so we also run 10-fold stratified cross-validation on all 90 launches.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

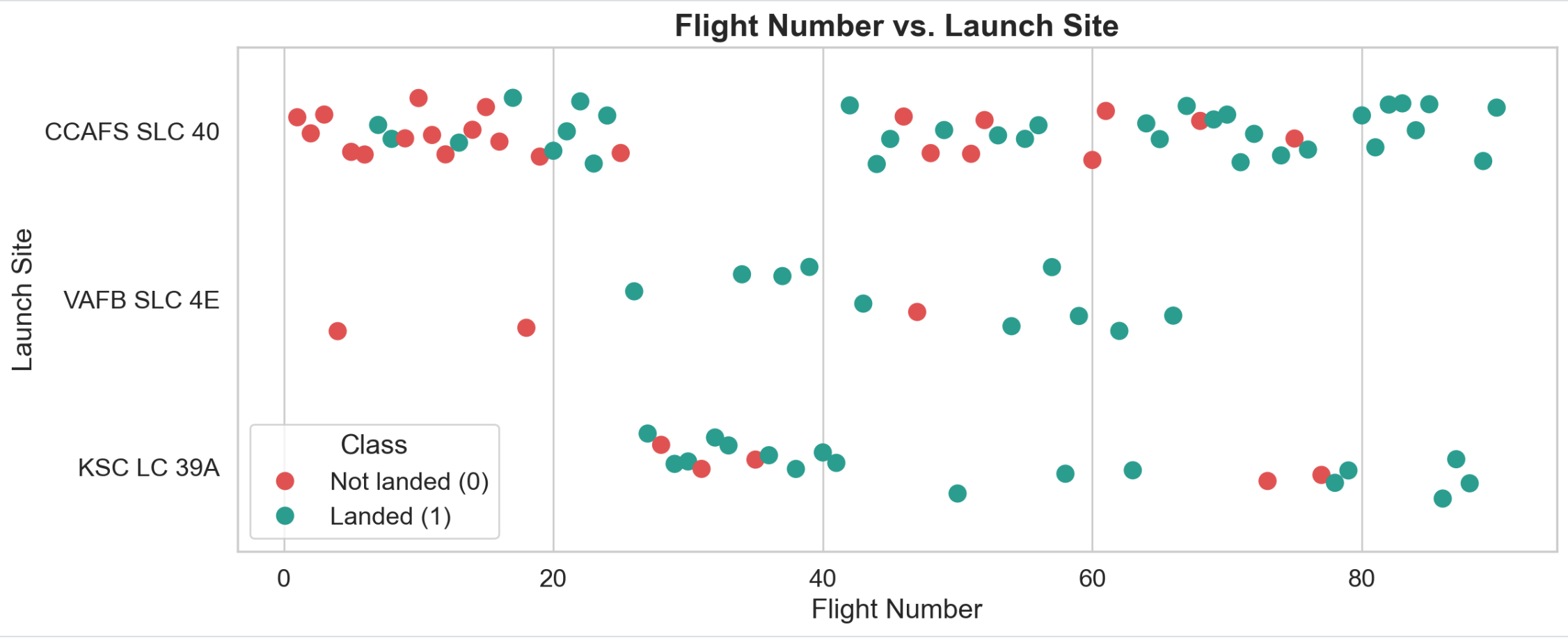
# Flight Number vs. Payload Mass




**Insight:** Later flights land more often (43% for flights 1–30 vs. 83% for flights 61–90). Heavy payloads (> 10,000 kg) still landed 13 of 15 times.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

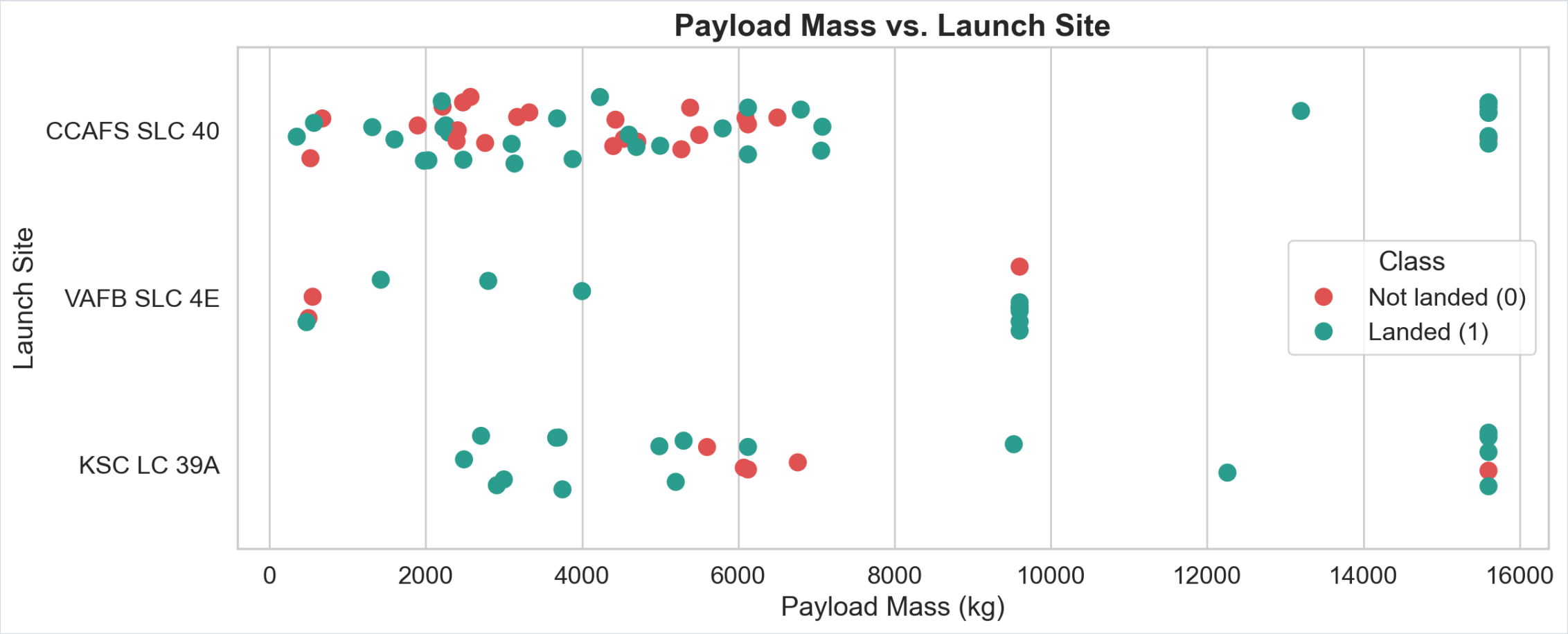
# Flight Number vs. Launch Site




**Insight:** CCAFS SLC 40 has the most launches (55) and the lowest rate (60%): its first 20 launches had 35% success, later ones 74%.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

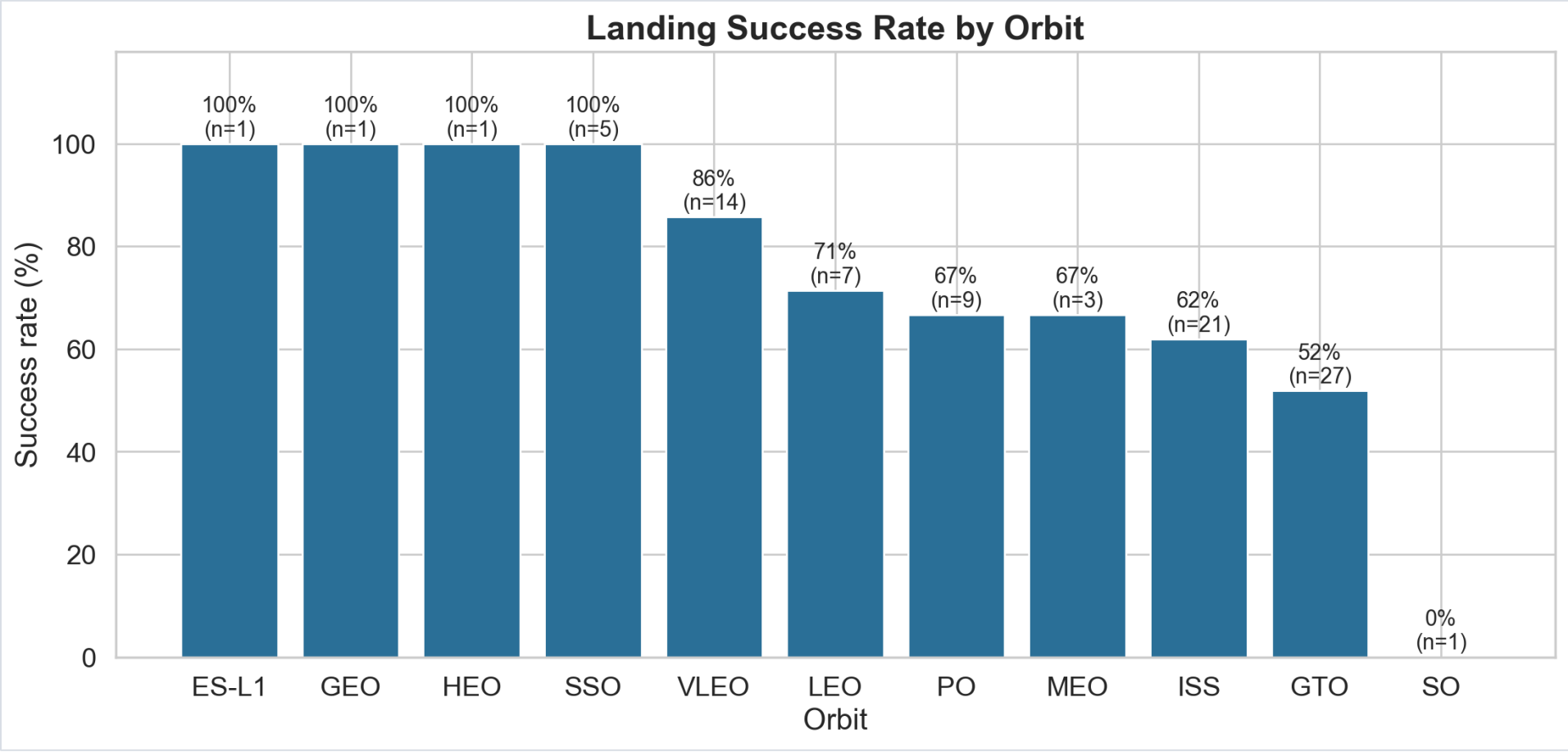
# Payload Mass vs. Launch Site




**Insight:** VAFB SLC 4E never launched more than 9,600 kg. All payloads above 10,000 kg flew from Florida, with high success.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

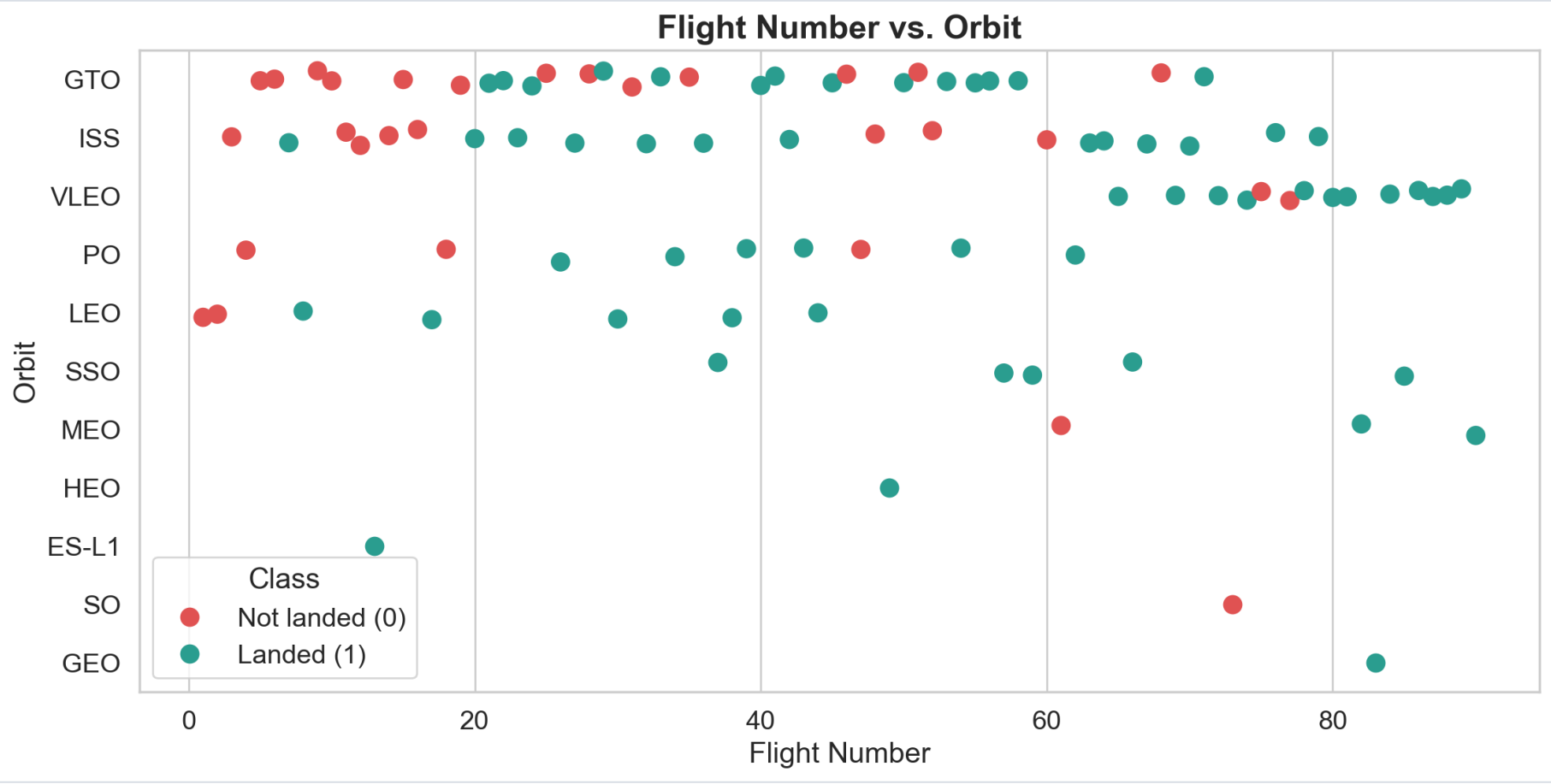
# Success Rate by Orbit Type




**Insight:** SSO, ES-L1, GEO and HEO show 100% but have very few launches. Among common orbits, VLEO is best (86%, n=14) and GTO is weakest (52%, n=27).

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

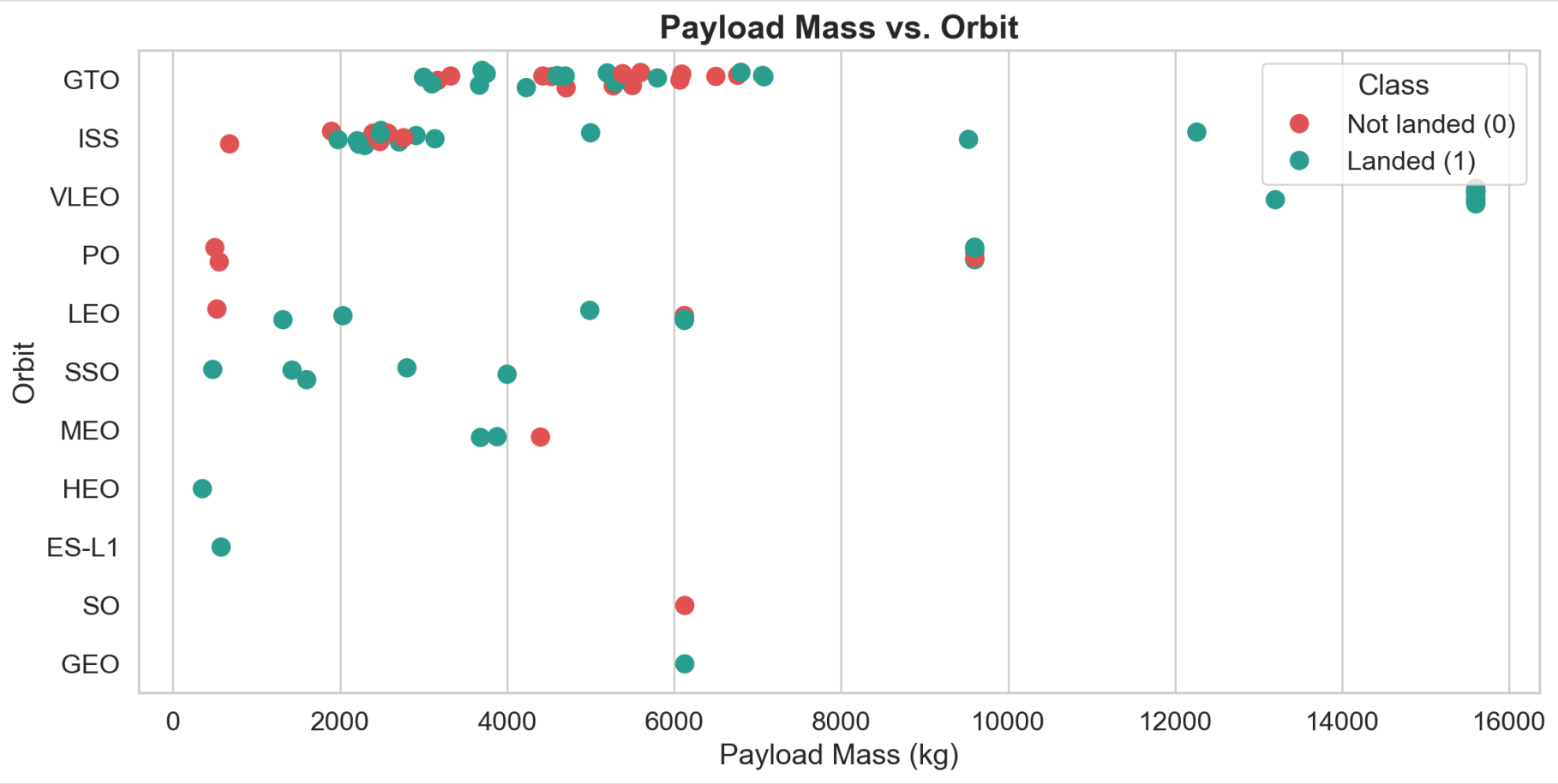
# Flight Number vs. Orbit




**Insight:** LEO failed only in the first two flights, then always landed. GTO shows no clear improvement (52%). VLEO flights all came late and landed 86%.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

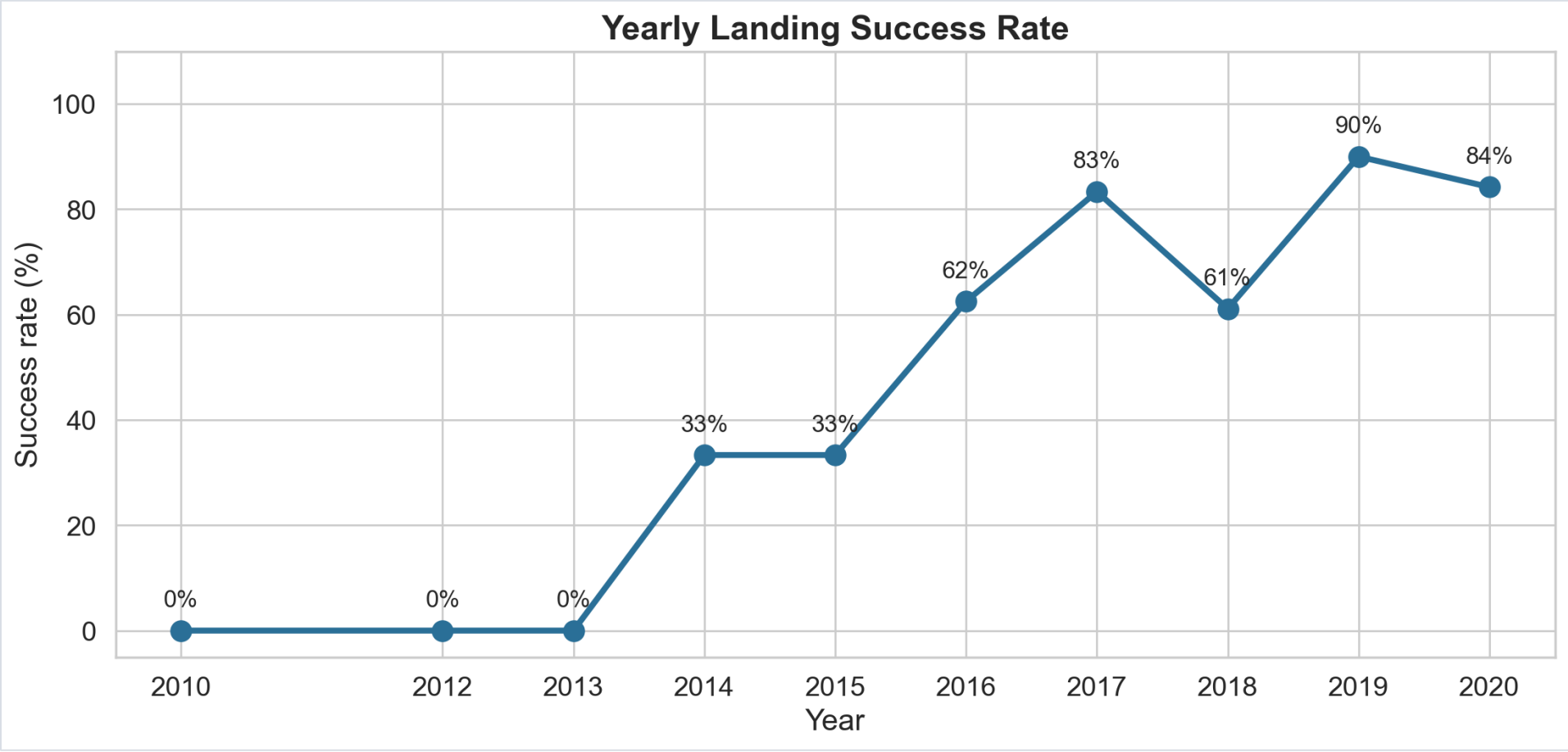
# Payload Mass vs. Orbit




**Insight:** Heavy VLEO payloads (13,200–15,600 kg) land well. In GTO, heavier payloads land less often (62% for ≤ 5,000 kg vs. 43% for > 5,000 kg).

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

# Launch Success Yearly Trend




**Insight:** Success rose from 0% (2010–2013) to 90% in 2019 (84% in 2020), with one dip in 2018 (61%).

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

# SQL Results: Launch Sites

```
SELECT DISTINCT Launch_Site FROM
SPACEXTABLE;
```

| Launch_Site |
|---|
| CCAFS LC-40 |
| VAFB SLC-4E |
| KSC LC-39A |
| CCAFS SLC-40 |

```
SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

| Date | Booster Version | Launch Site | Payload | PAYLOAD MASS KG | Orbit | Customer | Mission Outcome | Landing Outcome |
|---|---|---|---|---|---|---|---|---|
| 2010-06-04 | F9 v1.0 B0003 | CCAFS LC-40 | Dragon Spacecraft Qualific | 0 | LEO | SpaceX | Success | Failure (parachute) |
| 2010-12-08 | F9 v1.0 B0004 | CCAFS LC-40 | Dragon demo flight C1, two | 0 | LEO (ISS) | NASA (COTS) NRO | Success | Failure (parachute) |
| 2012-05-22 | F9 v1.0 B0005 | CCAFS LC-40 | Dragon demo flight C2 | 525 | LEO (ISS) | NASA (COTS) | Success | No attempt |
| 2012-10-08 | F9 v1.0 B0006 | CCAFS LC-40 | SpaceX CRS-1 | 500 | LEO (ISS) | NASA (CRS) | Success | No attempt |
| 2013-03-01 | F9 v1.0 B0007 | CCAFS LC-40 | SpaceX CRS-2 | 677 | LEO (ISS) | NASA (CRS) | Success | No attempt |

**Insight:** 4 unique sites; the first 5 'CCA' launches (2010–2013) were early Dragon / CRS missions with no landing attempt or parachute failures.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

# SQL Results: Payloads

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Total payload – NASA (CRS)

**45,596 kg**

```
SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

Average payload – F9 v1.1

**2,928.4 kg**

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ =
  (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

**12 boosters carried the maximum payload of 15,600 kg:**

| Booster | Booster |
|---|---|
| F9 B5 B1048.4 | F9 B5 B1049.5 |
| F9 B5 B1049.4 | F9 B5 B1060.2 |
| F9 B5 B1051.3 | F9 B5 B1058.3 |
| F9 B5 B1056.4 | F9 B5 B1051.6 |
| F9 B5 B1048.5 | F9 B5 B1060.3 |
| F9 B5 B1051.4 | F9 B5 B1049.7 |

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

# SQL Results: Success Rates

```
SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

| Mission_Outcome | Total |
|---|---|
| Failure (in flight) | 1 |
| Success | 98 |
| Success | 1 |
| Success (payload status unclear) | 1 |

**Grouped:** Success = 100, Failure = 1 → 99.0% mission success

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ <
6000;
```

| Booster Version | Payload kg |
|---|---|
| F9 FT B1022 | 4696 |
| F9 FT B1026 | 4600 |
| F9 FT  B1021.2 | 5300 |
| F9 FT  B1031.2 | 5200 |

**Insight:** Mission success is very high (100 of 101), but booster landing is the hard part. Only F9 FT boosters landed on a drone ship with 4,000–6,000 kg payloads.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

# SQL Results: Ranking Landing Outcomes

```
SELECT Landing_Outcome, COUNT(*) AS Total FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY Total DESC;
```

**Insight:** In the early years, 'No attempt' was most common (10). Drone-ship landings were 50/50 (5 success vs. 5 failure), and ground-pad landings already worked (3 of 3).

| Rank | Landing_Outcome | Total |
|---|---|---|
| 1 | No attempt | 10 |
| 2 | Success (drone ship) | 5 |
| 3 | Failure (drone ship) | 5 |
| 4 | Success (ground pad) | 3 |
| 5 | Controlled (ocean) | 3 |
| 6 | Uncontrolled (ocean) | 2 |
| 7 | Failure (parachute) | 2 |
| 8 | Precluded (drone ship) | 1 |

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

# SQL Results: Time Analysis

```
SELECT MIN(Date) FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

First successful ground-pad landing

## 22 December 2015

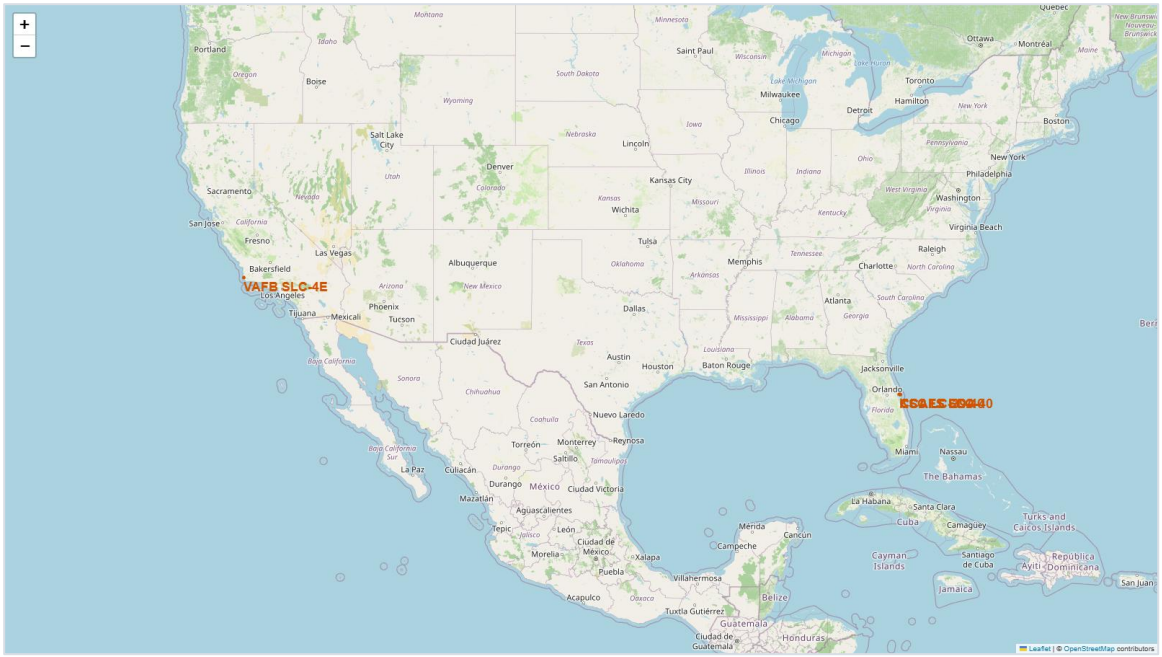
```
SELECT substr(Date,6,2) AS Month, Landing_Outcome,
       Booster_Version, Launch_Site FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
  AND substr(Date,1,4) = '2015';
```

| Month | Month name | Landing Outcome | Booster Version | Launch Site |
|---|---|---|---|---|
| 01 | January | Failure (drone ship) | F9 v1.1 B1012 | CCAFS LC-40 |
| 04 | April | Failure (drone ship) | F9 v1.1 B1015 | CCAFS LC-40 |

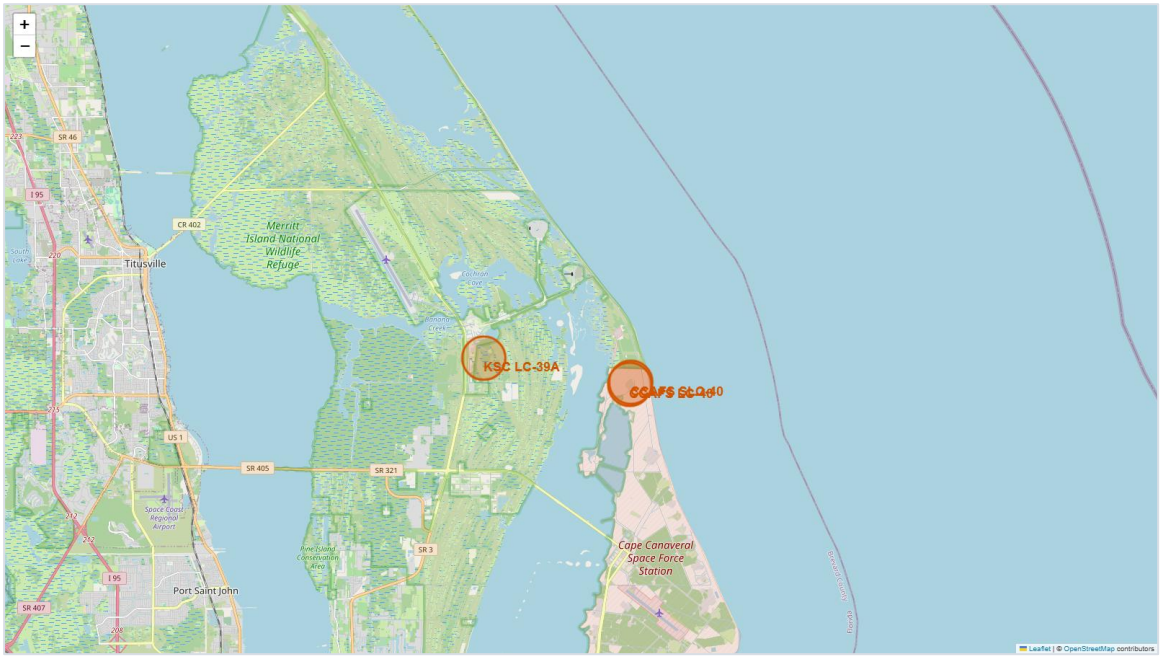
**Insight:** Both 2015 drone-ship failures were F9 v1.1 boosters from CCAFS LC-40 (January and April). Eight months later (Dec 2015) the first ground-pad landing succeeded.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

# Folium: All Launch Sites on the Map




All sites (USA)

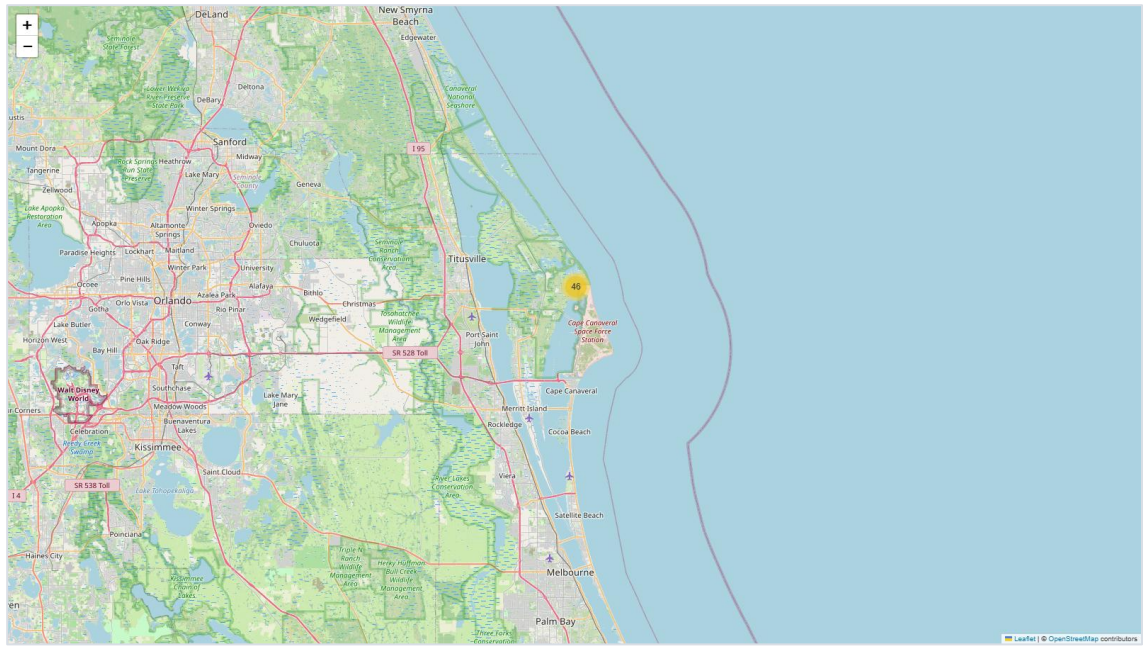



Cape Canaveral close-up (LC-40 and SLC-40 labels overlap: same pad area)

**Insight:** 3 sites in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and 1 in California (VAFB SLC-4E). All sit on the coast in the south of the USA – closer to the equator gives extra speed from Earth's rotation, and launches fly over the ocean.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

# Folium: Launch Outcomes per Site




MarkerCluster – number of launches per site

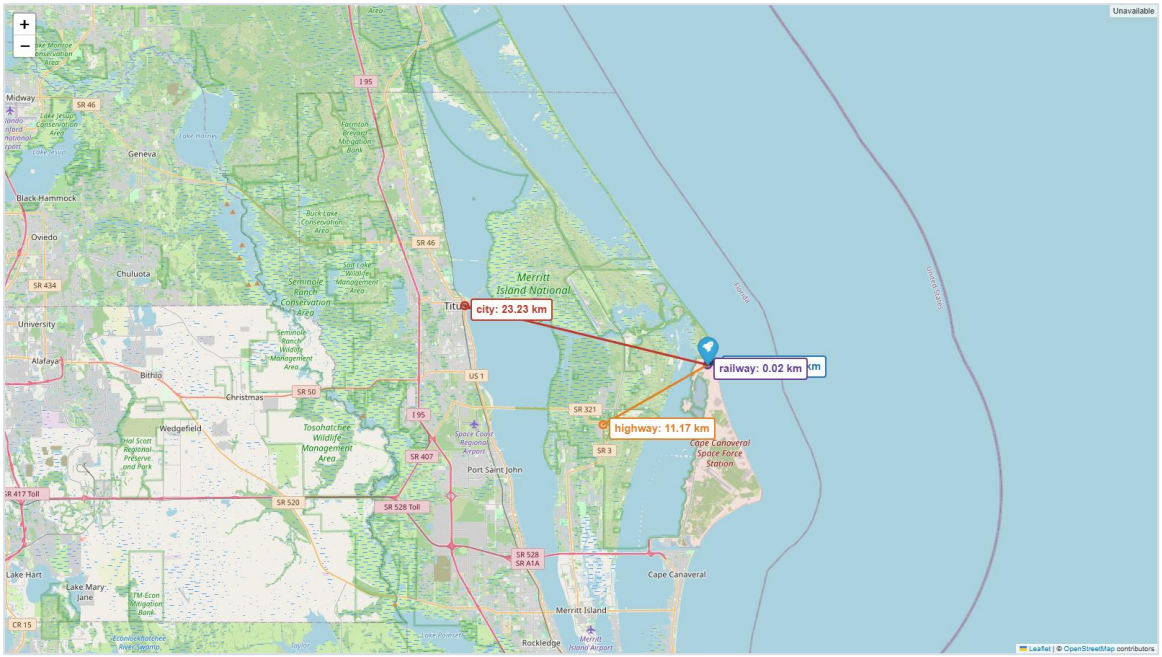



KSC LC-39A cluster opened – green = landed, red = not landed

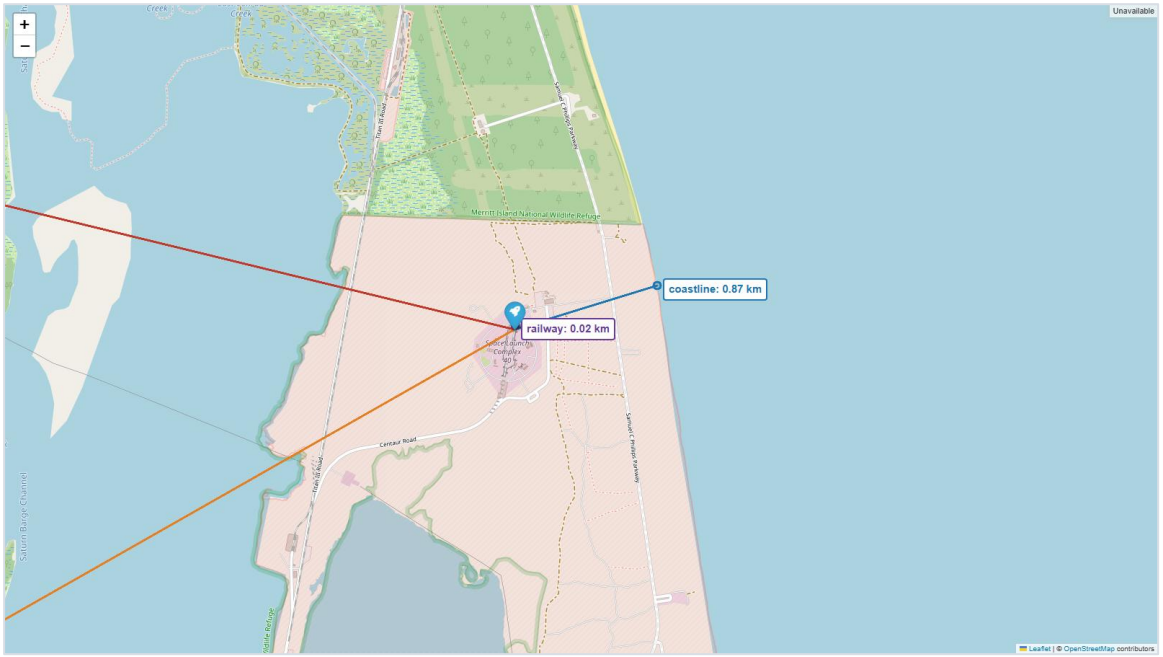
**Insight:** KSC LC-39A: 10/13 landed (77%) | CCAFS SLC-40: 3/7 landed (43%) | VAFB SLC-4E: 4/10 landed (40%) | CCAFS LC-40: 7/26 landed (27%)

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

# Folium: Proximity Analysis (CCAFS SLC-40)




Overview: highway and city




Close-up: coastline and railway

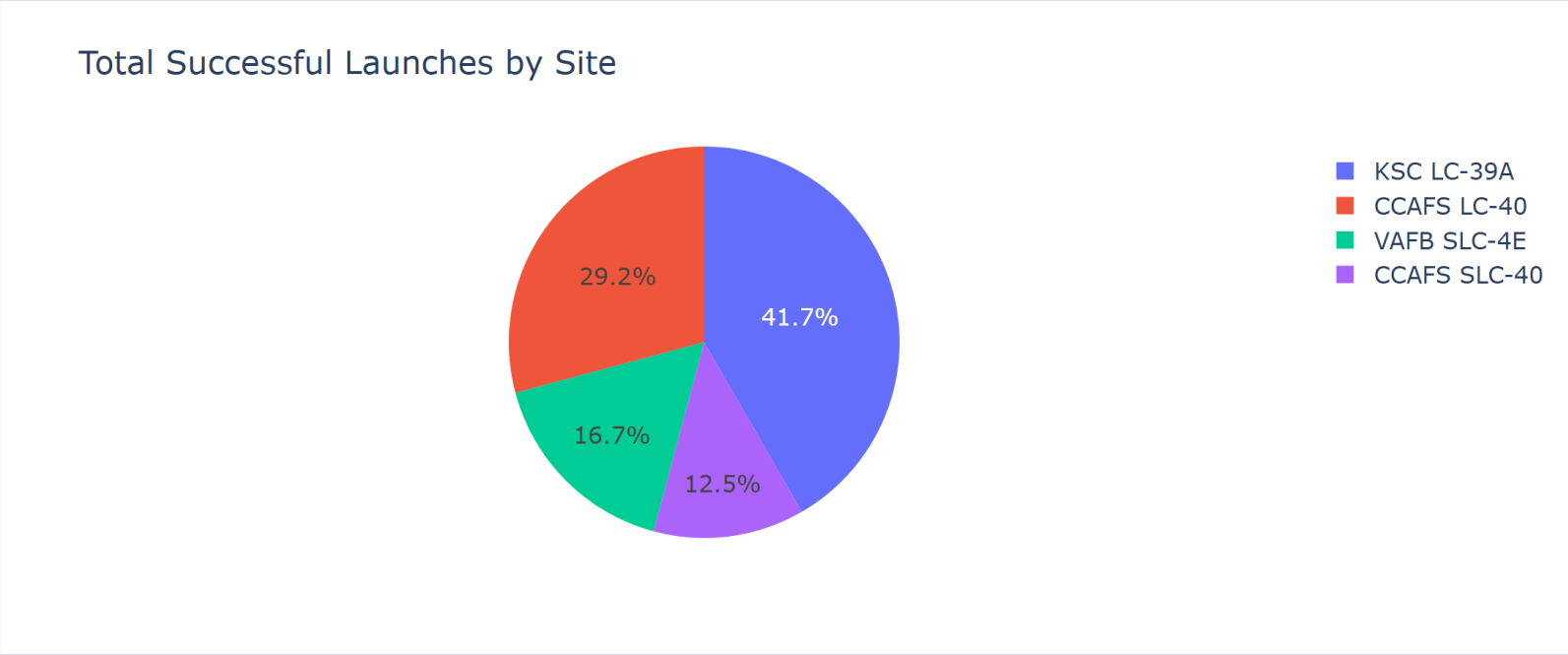
| Nearest | Name (OpenStreetMap) | Distance |
|---|---|---|
| Coastline | – | 0.87 km |
| Railway | – | 0.02 km |
| Highway | Space Commerce Way | 11.17 km |
| City | Titusville | 23.23 km |

## Findings

- Very close to the coast (0.87 km): a failed rocket falls into the ocean
- Very close to a railway (0.02 km): heavy parts arrive by train
- Highway 11.2 km away: road access for people and supplies
- Nearest city (Titusville) 23.2 km away: far from people for safety

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

# Dash: Successful Launches by Site (All Sites)

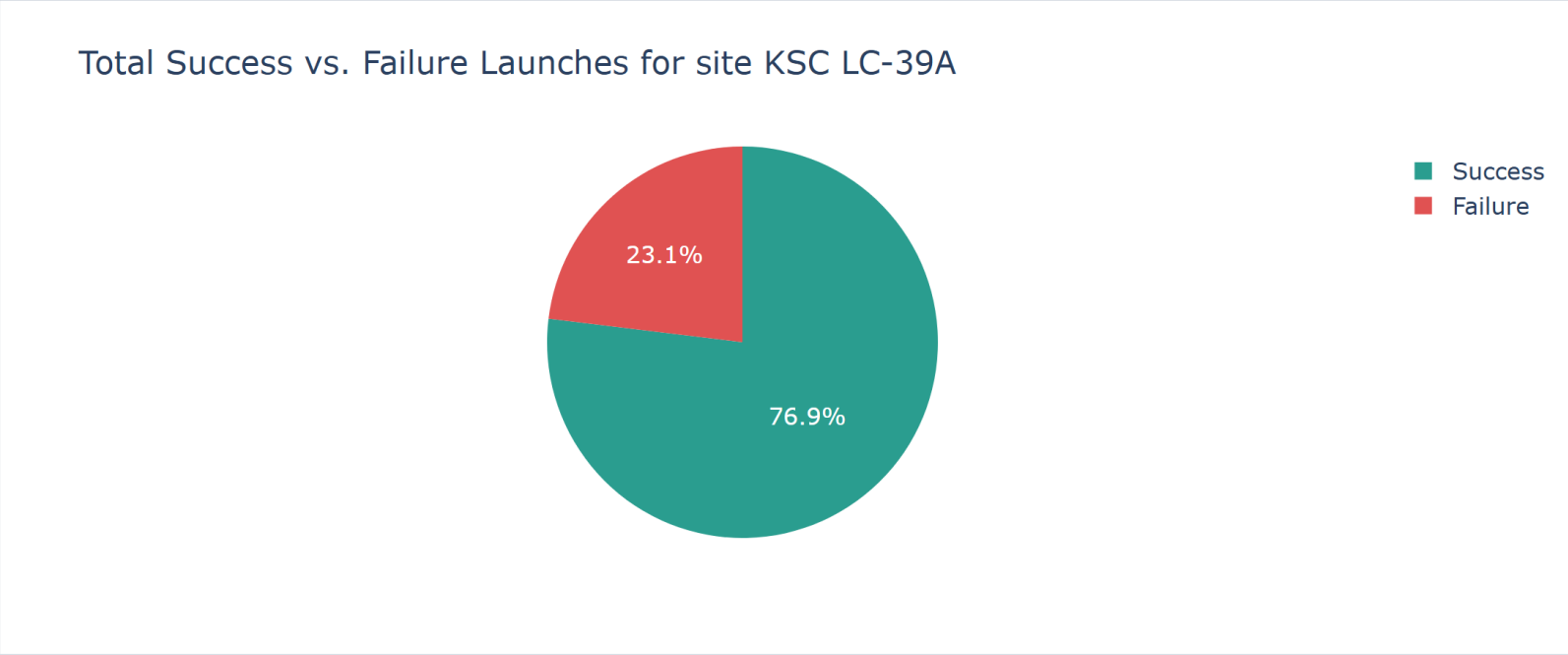



| Launch Site | Successes | Share |
|---|---|---|
| KSC LC-39A | 10 | 41.7% |
| CCAFS LC-40 | 7 | 29.2% |
| VAFB SLC-4E | 4 | 16.7% |
| CCAFS SLC-40 | 3 | 12.5% |

**Insight:** KSC LC-39A produced the largest share of all successful launches (41.7%); CCAFS SLC-40 the smallest (12.5%).

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/07_dash_app.py

# Dash: Highest Success Rate Site – KSC LC-39A

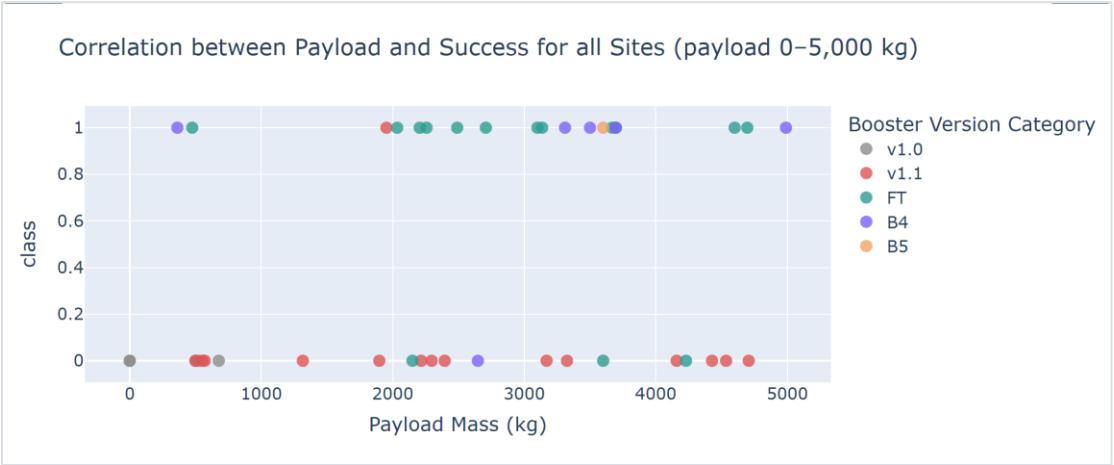


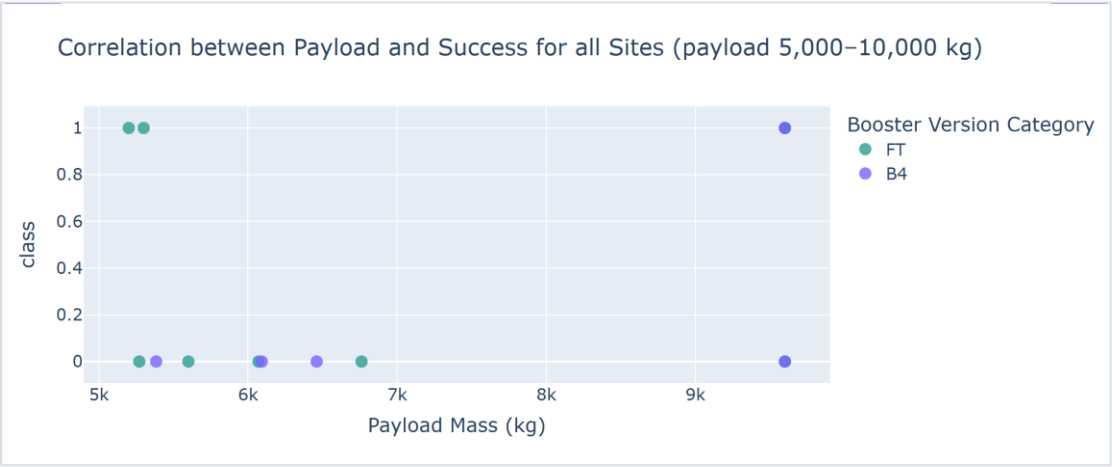

| Launch Site | Launches | Success rate |
|---|---|---|
| KSC LC-39A | 13 | 76.9% |
| CCAFS SLC-40 | 7 | 42.9% |
| VAFB SLC-4E | 10 | 40.0% |
| CCAFS LC-40 | 26 | 26.9% |

**Insight:** KSC LC-39A has the highest success rate: 10 of 13 launches (76.9% success, 23.1% failure).

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# Dash: Payload vs. Launch Outcome (All Sites)



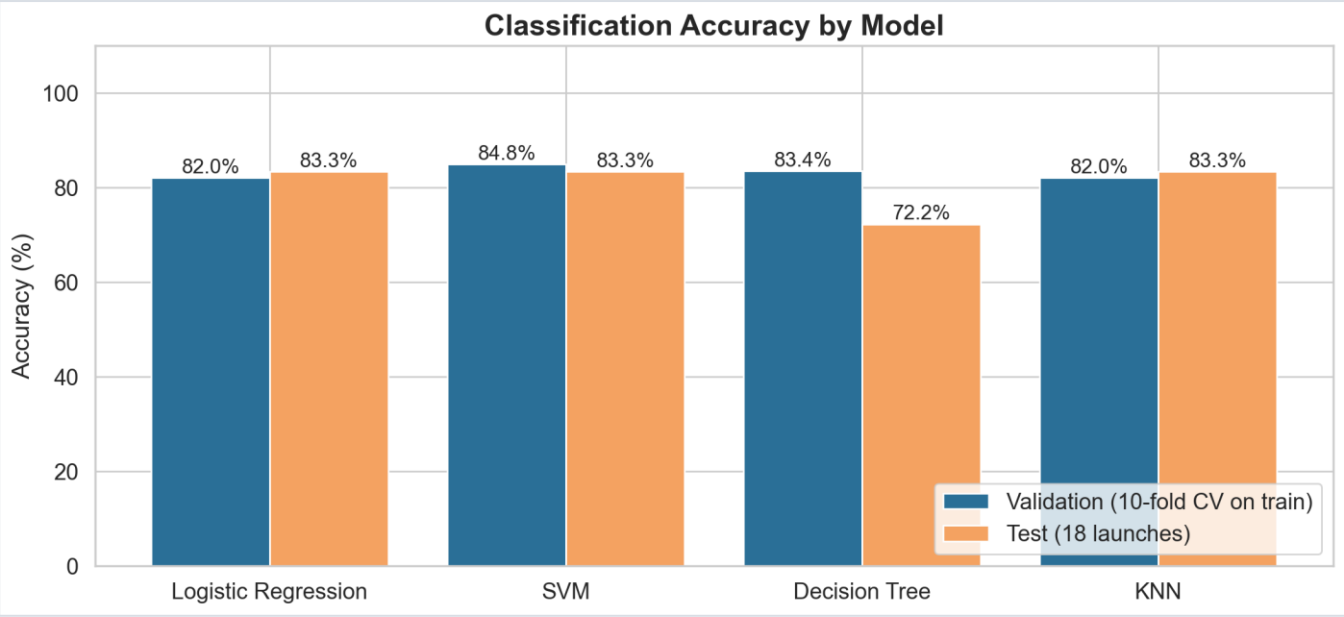

| Payload (kg) | Launches | Success |
|---|---|---|
| 0–2,000 | 13 | 23% |
| 2,000–4,000 | 21 | 62% |
| 4,000–6,000 | 13 | 38% |
| 6,000–8,000 | 4 | 0% |
| 8,000–10,000 | 5 | 60% |

| Booster | Launches | Success |
|---|---|---|
| B4 | 11 | 55% |
| B5 | 1 | 100% |
| FT | 24 | 67% |
| v1.0 | 5 | 0% |
| v1.1 | 15 | 7% |

**Insight:** Best payload range: 2,000–4,000 kg (62%, n = 21); worst: 6,000–8,000 kg (0 of 4). Best booster: FT (67%, n = 24); B5 is 1 of 1 (too few); v1.1 lowest (7%).

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/07_dash_app.py

# Classification Accuracy

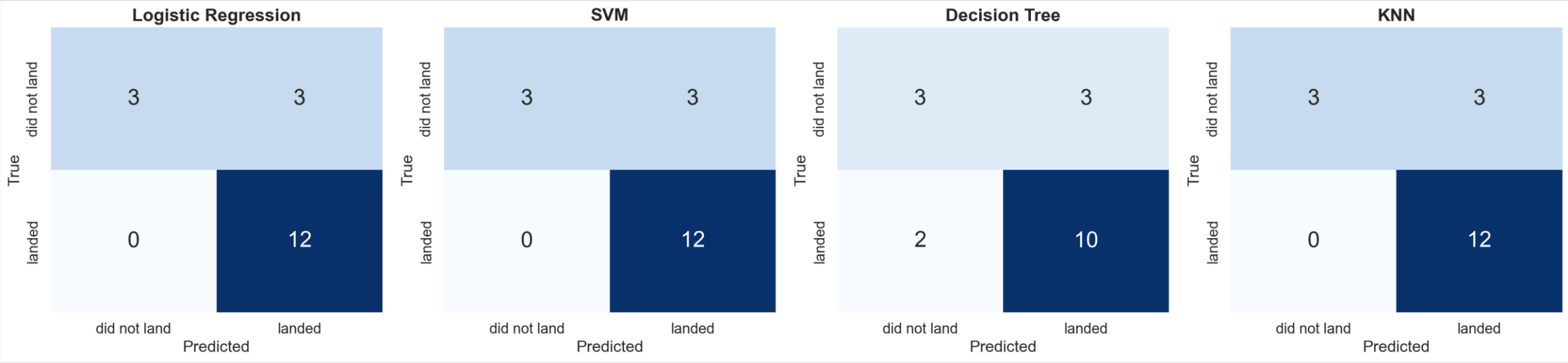



| Model | Validation | Test | 10-fold CV (all 90) |
|---|---|---|---|
| Logistic Regression | 82.0% | 83.3% | 83.3% ± 12.4 |
| SVM | 84.8% | 83.3% | 84.4% ± 10.2 |
| Decision Tree | 83.4% | 72.2% | 78.9% ± 13.6 |
| KNN | 82.0% | 83.3% | 81.1% ± 8.7 |

| Model | Best parameters (GridSearchCV) |
|---|---|
| Logistic Regression | C: 0.1, solver: lbfgs |
| SVM | C: 1, gamma: 0.0316, kernel: sigmoid |
| Decision Tree | criterion: gini, max_depth: 10, max_features: sqrt, min_samples_leaf: 1, min_samples_split: 2, splitter: random |
| KNN | algorithm: auto, n_neighbors: 7, p: 1 |

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

# Confusion Matrix




**LR, SVM and KNN:** same matrix – TP = 12, TN = 3, FP = 3, FN = 0. Every landed booster is found (recall = 100%), but 3 of 6 non-landings are wrongly predicted as landed.

**Decision Tree:** TP = 10, TN = 3, FP = 3, FN = 2 – it also misses 2 real landings.

> **Insight:** The main error is the false positive: the models are too optimistic about landing. For cost estimates this is the risky mistake (a bid would assume a reusable booster).

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

# Best Model

**Selected model**

## Support Vector Machine

Kernel: sigmoid, C = 1, gamma = 0.032

Validation accuracy: 84.8% (best of 4)

Test accuracy: 83.3% (tied best)

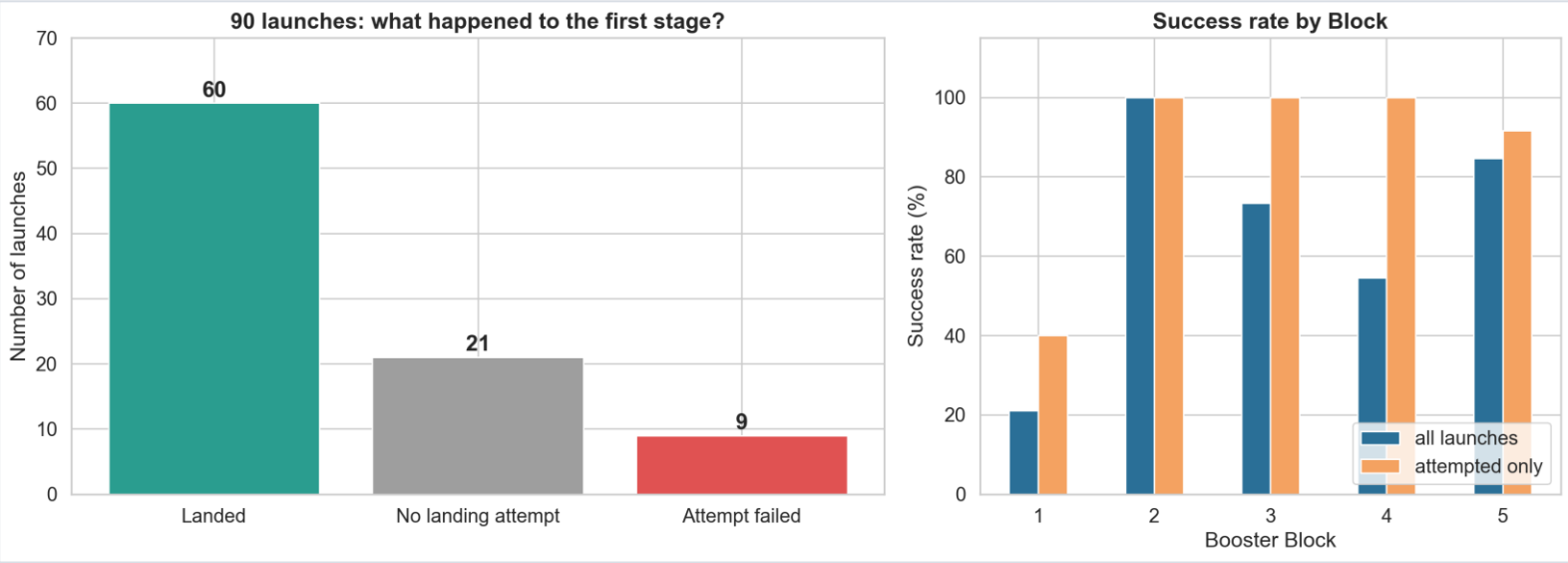
10-fold CV on all data: 84.4% (best of 4)

### Why SVM – and an honest warning

- Logistic Regression, SVM, KNN all score 83.3% on the test set: the same 15 of 18 correct.
- With only 18 test launches, one prediction = 5.6 points, so the test set cannot separate these models.
- SVM is highest on both validation (84.8%) and 10-fold CV on all 90 launches (84.4%).
- But the CV spread is about ±10 points, so the gap to Logistic Regression (83.3%) is small.
- Logistic Regression is a good simpler alternative.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

# What Really Drives Landing Success?




**Class = 0: 30 launches**

- No landing attempt: 21
- Attempt failed: 9

**Attempted landings: 60 of 69 succeeded (87.0%)**

Attempted, by Block: 1 = 40%, 2–4 = 100%, 5 = 92%

Reused booster: 78% vs. new: 58%

**Insight:** Most 'failures' are planned: 21 of 30 Class-0 launches had no landing attempt (12 of them GTO). Once SpaceX tries, it lands 87% of the time – so predicting the mission plan (orbit, payload) matters more than the landing itself.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

# Conclusion

- Experience matters: success rose from 0% (2010–2013) to 90% (2019); flights 61–90 land 83% vs. 43% for flights 1–30.
- Orbit matters: VLEO 86%, ISS 62%, GTO only 52% (heavy GTO payloads are hardest).
- Launch sites are all next to the coast, near rail, and far from cities; KSC LC-39A has the best success rate in the dashboard data (77%).
- Payloads of 2,000–4,000 kg and FT boosters show the best success in the dashboard data.
- LR, SVM and KNN reach 83.3% test accuracy; SVM is chosen (best validation / CV), but the small test set means the models are nearly equal.

## Creative insight

"Failure" is mostly a choice. 21 of 30 non-landings had no landing attempt; 12 of them were GTO missions.

This also explains the 2018 dip: 6 launches that year did not try to land.

When SpaceX tries, it lands 87% of the time.

Advice for Space Y: price a bid in two steps – (1) will SpaceX try to land for this orbit and payload? (2) if yes, expect the booster to be reused.

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone

# Appendix: Data Sources and Notes

- SpaceX REST API v4 (api.spacexdata.com) – live server returned HTTP 522/525 in Oct 2026; used IBM static copy of /launches/past and archived API responses from web.archive.org (rockets, payloads, launchpads, cores).
- Validation: our dataset_part_1.csv matches IBM's reference in 15 of 17 columns; small differences in PayloadMass and ReusedCount come from a newer API snapshot.
- Wikipedia: List of Falcon 9 and Falcon Heavy launches, revision oldid = 1027686922.
- IBM course files: Spacex.csv (SQL), spacex_launch_geo.csv (Folium), spacex_launch_dash.csv (Dash).
- OpenStreetMap via Overpass API for nearest coastline, railway, highway and city (© OpenStreetMap contributors, ODbL). Note: the course file places KSC LC-39A slightly inland, so we show proximity for CCAFS SLC-40.
- Tools: Python 3.12, pandas, numpy, requests, BeautifulSoup, matplotlib, seaborn, sqlite3, folium, plotly / Dash, scikit-learn, Playwright (screenshots).
- Maps (interactive HTML): images/folium_1_launch_sites.html, folium_2_launch_outcomes.html, folium_3_proximity.html

**GitHub:** https://github.com/khoalee252006/spacex-falcon9-capstone